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#101#

well Done!

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Midsemester Exam #1

Only one sheet of paper 8 1/2 x 11 is allowed

No phones or i-pads or computers are allowed

Duration: 70 minutes --- September 23, 2025

All Problems must be done

Problem 1: (45 points)

Determine the conduction heat flux through various plane layers that are subjected to boundary temperatures of $T_{s,1} = 310$ K and $T_{s,2} = 290$ K at atmospheric pressure.

(a) Case A: The plane layer is aluminum. Determine the heat flux q''_x for $L_{tot} = 400$ μm and $L_{tot} = 400$ nm.

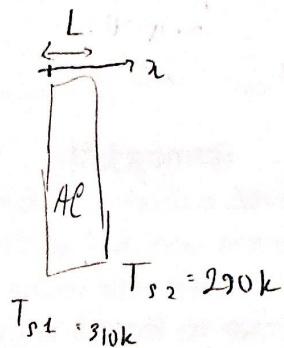
(b) Case B: Conduction occurs through an air layer. Determine the heat flux q''_x for $L_{tot} = 400$ μm and $L_{tot} = 400$ nm.

(c) The composite is composed of air held between two aluminum sheets. Determine the heat flux q''_x for $L_{tot} = 400$ μm (with aluminum sheet thickness of $\delta = 40$ μm) and $L_{tot} = 400$ nm (with aluminum sheet thickness of $\delta = 40$ nm)

Please note that $k_{air} = 0.0263$ W/(m·K) and $k_{Al} = 237$ W/(m·K)

Q. 1:

a) ✓



$$k_{Al} = 237 \frac{W}{m \cdot K}$$

$$\frac{W}{m \cdot K} \cdot \frac{K}{m} = \frac{W}{m^2}$$

$$q''_x = -k_{Al} \frac{T_{s2} - T_{s1}}{L}$$

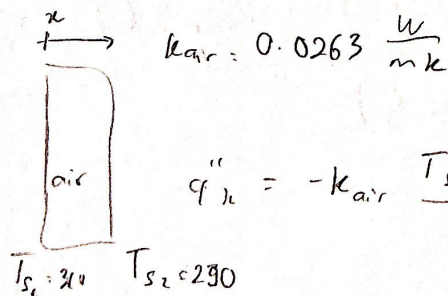
For $L = 400 \mu m = 400 \times 10^{-6} m = 4 \times 10^{-4} m$:

$$q''_x = -237 \frac{290 - 310}{4 \times 10^{-4}} = \boxed{1.185 \times 10^7 W/m^2} \checkmark$$

For $L = 400 nm = 400 \times 10^{-9} m = 4 \times 10^{-7} m$ ✓

$$q''_x = -237 \frac{290 - 310}{4 \times 10^{-7}} = \boxed{1.185 \times 10^{10} W/m^2} \checkmark \quad \text{✓}$$

b) ✓



$$k_{air} = 0.0263 \frac{W}{m \cdot K}$$

$$q''_x = -k_{air} \frac{T_{s2} - T_{s1}}{L}$$

For $L = 400 \mu m = 4 \times 10^{-4} m$:

$$q''_x = -0.0263 \frac{290 - 310}{4 \times 10^{-4}} = \boxed{1315 W/m^2} \checkmark$$

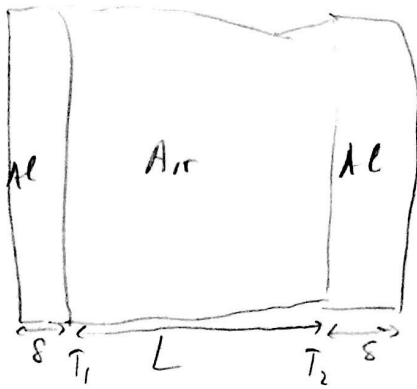
For $L = 400 nm = 4 \times 10^{-7} m$:

$$q''_x = -0.0263 \frac{290 - 310}{4 \times 10^{-7}} = \boxed{1315 \times 10^3 W/m^2}$$

c)

$$T_{s1} = 310 \text{ K}$$

$$T_{s2} = 290 \text{ K}$$



$$k_{air} = 0.0263 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

$$k_{Al} = 237 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

Flux:

$$-k_{Al} \frac{T_1 - T_{s1}}{\delta} + k_{air} \frac{T_2 - T_1}{L} - k_{Al} \frac{T_{s2} - T_2}{\delta}$$

$$= -k_{Al} \frac{T_1 - T_{s1} + T_{s2} - T_2}{\delta} - k_{air} \frac{T_{s2} - T_{s1}}{L}$$

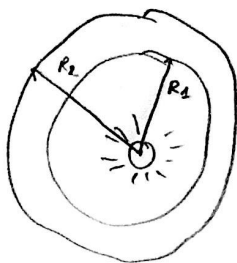
$$= \frac{-k_{Al} L (T_1 - T_{s1} + T_{s2} - T_2) - k_{air} \delta (T_{s2} - T_{s1})}{\delta L}$$

Problem 2: (30 points)

A 60-W light bulb is lit inside a spherical cell made of an opaque solid with unknown thermal conductivity. The inner and outer radii are 10 cm and 15 cm, respectively. By measuring the steady state temperatures at the inner and outer surfaces of the cell, we can determine the thermal conductivity of the material. In this case, the measured temperatures are $T = 200^\circ\text{C}$ and $T = 50^\circ\text{C}$. Determine the unknown thermal conductivity of the material.

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Assumptions:
- constant thermal conductivity
- T is dependent on r only



$$Q = 60\text{ W}$$

$$R_1 = 10\text{ cm} = 0.10\text{ m}$$

$$R_2 = 15\text{ cm} = 0.15\text{ m}$$

$$T_1 = 200^\circ\text{C} = 473\text{ K}$$

$$T_2 = 50^\circ\text{C} = 323\text{ K}$$

Balance equation ($r \in [0.10, 0.15]$)

$$\frac{k}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) + \frac{Q}{V_{\text{cell}}} = 0 \Rightarrow \frac{k}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) + \frac{60\text{ W}}{9.95 \times 10^{-3}\text{ m}^3} = 0$$

Unit: $\frac{\text{W}}{\text{m}^2\text{K}} \cdot \frac{1}{\text{m}^2} \cdot \frac{1}{\text{m}} \cdot \text{m}^2 \cdot \frac{\text{K}}{\text{m}} = \frac{\text{W}}{\text{m}^3}$

Boundary conditions:

$$T(r = 0.1) = 473\text{ K}$$

$$T(r = 0.15) = 323\text{ K}$$

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = \frac{-6031}{k} r^2$$

$$r^2 \frac{dT}{dr} = \frac{-2010}{k} r^3 + A$$

$$\frac{dT}{dr} = \frac{-2010}{k} r + \frac{A}{r^2}$$

$$T = \frac{-1005}{k} r^2 - \frac{A}{r} + B$$

$$T(r) = -\frac{1005}{k} r^2 - \frac{A}{r} + B$$

$$B.C: T(0.10) = 473 \text{ K}$$

$$\therefore 473 = -\frac{1005}{k} (0.10)^2 - \frac{A}{0.10} + B \quad (1)$$

$$B.C: T(0.15) = 393 \text{ K}$$

$$\therefore 393 = -\frac{1005}{k} (0.15)^2 - \frac{A}{0.15} + B \quad (2)$$

Solve the linear system (1), (2) to yield:

$$A = -\frac{45(k - 0.08375)}{k}$$

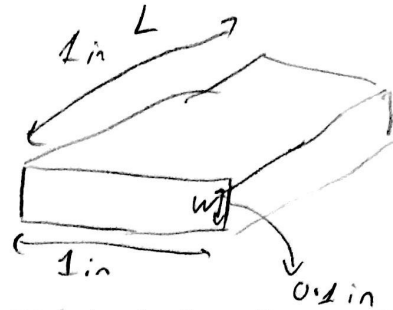
$$B = \frac{23(k + 2.076)}{k}$$

$$T(r) = -\frac{1005}{k} r^2 + \frac{45(k - 0.08375)}{k} \frac{1}{r} + \frac{23(k + 2.076)}{k}$$

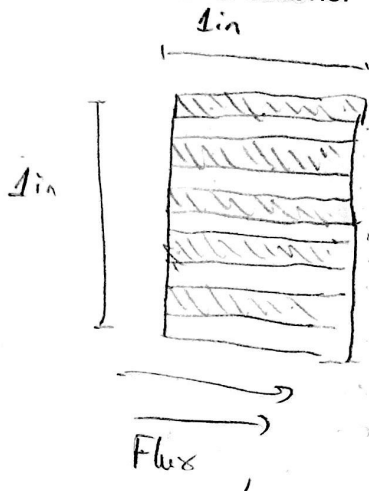
$$T(0.10) = 473 = -\frac{1005}{k} (0.10)^2 + \frac{45(k - 0.08375)}{k} \frac{1}{0.10} + \frac{23(k + 2.076)}{k}$$

$$\therefore k \approx 0.16 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

Problem 3: (30 points)



A cube 1.0-in. long on each side, is constructed of equal-sized, alternating parallel layers of copper and Pyrex glass. There are five layers of copper and five layers of glass. A question has risen whether this cube would be a better conductor of heat (for a fixed overall ΔT) if it were oriented so that heat was conducted parallel to the alternating layers or normal to them. Find the ratio of the heat conduction rates for the two different orientations.



Assumptions: - 1D
- constant k

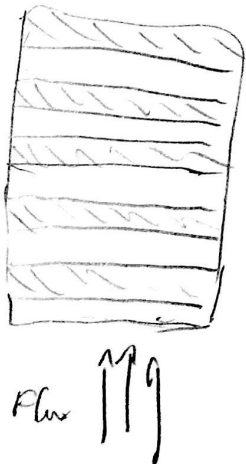
Copper $k_c \equiv \text{thermal conductivity copper}$
Pyrex $k_p \equiv \text{thermal conductivity pyrex glass}$

Heat conduction rate parallel $= q_c + q_p$

$$q_{||} = -5k_c A_c \frac{\Delta T}{L_c} - 5k_p A_p \frac{\Delta T}{L_p} \checkmark$$

$$= -5k_c \times 0.1 \frac{\Delta T}{1} - 5k_p \times 0.1 \frac{\Delta T}{1}$$

$$= -0.5k_c \Delta T - 0.5k_p \Delta T$$



Heat conduction parallel:

$$q_{\perp} = -5k_c A_c \frac{\Delta T}{w} - 5k_p A_p \frac{\Delta T}{w}$$

$$= -5k_c \times 1 \cdot \frac{\Delta T}{0.1} - 5k_p \times 1 \cdot \frac{\Delta T}{0.1}$$

$$= -50k_c \Delta T - 50k_p \Delta T$$

$$\frac{q_{\perp}}{q_{||}} = \frac{-50k_c \Delta T - 50k_p \Delta T}{-0.5k_c \Delta T - 0.5k_p \Delta T} = \boxed{100} \sim$$